INCH-POUND
MIL-I-60525A
30 March 1990
SUPERSEDING
MIL-I-60525
29 July 1966 and
AMENDMENT 2
21 May 1987

MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M99 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M99 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CAGE Code 19200 and 19203)

10520406	Initiator, Cartridge Actuated, M99, Metal Parts Assembly
8830984	Box, Fiberboard (Inner) Packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107
8830985	Box, Packing, Ammunition for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107
9241155	Box, Fiberboard (Outer) Alternative, Packing Ammunition, for Initiator, Cartridge Actuated M27, M86, M87, M98, M99, or XM107

U.S. AIR FORCE

TPO 00-845-1059	Initiator, Cartridge Actuated, M99

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5000.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the initiators shall conform strictly to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.4.1).

3.5 Firing pin release. The firing pin of the metal parts assembly, less safety pin, shall release from the cocked position when the application of an axial load of 18 to 25 pounds has withdrawn the initiator pin 0.370 inch, minimum, and 0.425 inch, maximum.

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3.6 Initiator pin security. Application of an axial load of 200 to 220 pounds to the initiator pin of the metal parts assembly shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 5 degrees in either direction using procedures in 4.6.6.

3.10 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspection. Inspection shall be in accordance with MIL-A-2550.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 20 to 25 °C (68 to 77 °F)
b. Altitude: Sea level
c. Humidity: 50 percent relative, maximum.
d. Atmospheric pressure: 14.7 psia

4.4 First article inspection. At the beginning of regular production, the contractor shall submit a first article sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 9 sets of unassembled parts, and 1 set of unassembled parts without plating or final protective finish. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.5.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.5.2 Examination. One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed. Four power (4×) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
1	Absence of output port	Visual

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Characteristic	Defect	Method of Inspection
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Pitch diameter of internal thread, max	Gage
106	Minor diameter of internal thread, max	Gage
107	Concentricity of datum location with pitch diameter of small external thread	Gage
108	Concentricity of pitch diameter of large thread with cavity	Gage
109	Concentricity of large external thread pilot with large external thread	Gage
110	Concentricity of counterbore with cavity, including perpendicularity with base of counterbore	Gage
111	Diameter of counterbore	Gage
112	Diameter of large thread pilot	Gage
113	Depth of cavity	Gage
114	Length to datum location	Gage
115	Thickness of wall, min	Gage
116	Metal defective	Visual
117	Protective finish inadequate or defective	Visual
118	Surface finish improper in cavity and angle of gas port	Visual
119	Burr	Visual
120	Foreign matter present	Visual
Minor		
201	Pitch diameter of large external thread, max	Gage
202	Pitch diameter of small external thread, max	Gage
203	Pitch diameter of internal thread, min	Gage
204	Diameter of small external thread relief	Gage
205	Diameter of small external thread pilot	Gage
206	Diameter of hole in gas port	Gage
207	Diameter of cavity	Gage
208	Depth of counterbore	Gage
209	Length of gas port	Gage
210	Length of small thread relief	Gage
211	Length from face of gas port to small thread	Gage
212	Length of external thread pilot	Gage
213	Thickness of wall, max	Gage
214	Depth of threaded hole	Measure
215	Diameter of chamfer at internal thread	Measure
216	Total length	Measure
217	Length of large external thread	Measure
218	Marking missing, incorrect, or illegible	Visual
219	Radius or chamfer missing or incorrect	Visual
220	Surface finish improper, except as otherwise classified	Visual

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4.5.2.2 Cap, Initiator, Drawing 10520408, covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of small internal thread, max	Gage
104	Minor diameter of small internal thread, max	Gage
105	Pitch diameter of large internal thread, max	Gage
106	Minor diameter of large internal thread, max	Gage
107	Concentricity of pitch diameter of large internal thread with midsection cavity, including perpendicularity with base of counterbore of large cavity	Gage
108	Concentricity of large cavity behind undercut with pitch diameter of large internal thread	Gage
109	Concentricity of pitch diameter of small internal thread with midsection cavity	Gage
110	Concentricity of hole between small cavity and boss cavity with midsection cavity	Gage
111	Concentricity of undercut in large cavity with large internal thread	Gage
112	Concentricity of small cavity with hole between small cavity and boss cavity	Gage
113	Concentricity of boss cavity with midsection cavity	Gage
114	Symmetry of width across flats with pitch diameter of external thread	Gage
115	Diameter of large cavity rear of undercut	Gage
116	Diameter of small cavity	Gage
117	Diameter of boss cavity, max	Gage
118	Diameter of hole between small cavity and boss cavity, max	Gage
119	Diameter of undercut at large internal thread	Gage
120	Diameter of safety pin holes, max	Gage
121	Length of effective exterior thread, min	Gage
122	Length of effective small interior thread, min	Gage
123	Length of exterior thread, max	Gage
124	Length of small interior thread, max	Gage
125	Width of undercut behind large interior thread	Gage
126	Location of safety pin hole	Gage
127	Height of boss	Gage
128	Metal defective	Visual
129	Protective finish inadequate or defective	Visual
130	Surface finish of undercut behind large interior thread improper	Visual
131	Burr	Visual
132	Foreign matter present	Visual

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Characteristic	Defect	Method of Inspection
Minor		
201	Pitch diameter of set screw thread	Gage
202	Minor diameter of set screw thread, max	Gage
203	Pitch diameter of external thread, max	Gage
204	Pitch diameter of small internal thread, min	Gage
205	Pitch diameter of large internal thread, min	Gage
206	Diameter of body, small end	Gage
207	Diameter of body, large end	Gage
208	Diameter of groove, small end	Gage
209	Diameter of groove on internal boss	Gage
210	Diameter of chamfer, large internal thread	Gage
211	Diameter of chamfer, small internal thread	Gage
212	Diameter of midsection cavity	Gage
213	Diameter of boss cavity, min	Gage
214	Diameter of hole between small cavity and boss cavity, min	Gage
215	Diameter of safety pin hole, min	Gage
216	Depth from base of large cavity rear of undercut to base of midsection cavity	Gage
217	Depth from base of large cavity rear of undercut to face of internal boss	Gage
218	Depth to bottom of large cavity rear of undercut	Gage
219	Depth of large cavity rear of undercut	Gage
220	Depth from base of large cavity rear of undercut to face of internal boss	Gage
221	Total length	Gage
222	Length of body, large end	Gage
223	Length of body, large end, including shoulder	Gage
224	Length of flat	Gage
225	Width of groove, small end	Gage
226	Width across flat	Gage
227	Location of set screw holes	Gage
228	Location of groove, small end	Gage
229	Safety pin hole obstructed	Visual
230	Marking missing, incorrect, or illegible	Visual
231	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
232	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Pin, Initiator, Drawing 10520409, covering a detail of Drawing 10520406.

Critical

None defined

Major		
101	Concentricity of groove in head with body	Gage
102	Concentricity of head with large body diameter	Gage
103	Concentricity of small body diameter with large body diameter	Gage
104	Large body diameter	Gage

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Characteristic	Defect	Method of Inspection
105	Small body diameter	Gage
106	Diameter of groove in body, min	Gage
107	Diameter of head	Gage
108	Length from groove in body to head	Gage
109	Length from head to hole	Gage
110	Location of groove in head	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of face of head, small body diameter, and groove in body, improper	Visual
114	Burr	Visual
115	Foreign matter present	Visual
Minor		
201	Concentricity of edges of flat end with large body	Gage
202	Symmetry of sides of flat end with large body diameter	Gage
203	Symmetry of hole with edges of flat end	Gage
204	Diameter of flat end	Gage
205	Diameter of undercut, min, if applicable	Gage
206	Diameter of groove in head	Gage
207	Diameter of groove in body, max	Gage
208	Diameter of hole	Gage
209	Length to groove in body	Gage
210	Length of small body near end	Gage
211	Length of large body diameter, including chamfer	Gage
212	Total length	Gage
213	Thickness of flat end	Gage
214	Radius or chamfer missing or incorrect	Visual
215	Surface finish improper except as otherwise classified	Visual

4.5.2.4 Seal, Drawing 10520415, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Diameter, outside, min	Gage
102	Width, min	Gage
103	Thickness, min	Gage
104	Cut, cracked, or split	Visual
105	Flash	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter, outside, max	Gage
202	Width, max	Gage
203	Thickness, max	Gage

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4.5.2.5 Housing, Firing Pin, Drawing 10520410, covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
1	Diameter of striker hole, min	Gage
Major		
101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of striker hole with cavity	Gage
104	Concentricity of cavity with body	Gage
105	Concentricity of pitch diameter of thread with body	Gage
106	Perpendicularity of face with body	Gage
107	Diameter of body, min	Gage
108	Diameter of cavity	Gage
109	Diameter of flange, min	Gage
110	Diameter of boss	Gage
111	Diameter of undercut, min	Gage
112	Total length, max	Gage
113	Width of undercut	Gage
114	Thickness of striker hole	Gage
115	Location of thread	Gage
116	Chamfer at striker hole	Gage
117	Height of boss	Gage
118	Metal defective	Visual
119	Protective finish, inadequate or defective	Visual
120	Surface finish of cavity, improper	Visual
121	Burr	Visual
122	Foreign matter present	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of body, max	Gage
203	Diameter of striker hole, max	Gage
204	Diameter of flange, max	Gage
205	Diameter of wrench holes	Gage
206	Depth of wrench holes	Gage
207	Total length, min	Gage
208	Length of body to flange	Gage
209	Location of wrench holes	Gage
210	Marking missing, incorrect or illegible	Visual
211	Radius or chamfer missing or incorrect, except as otherwise classified	Visual
212	Surface finish improper, except as otherwise classified	Visual

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4.5.2.6 Pin, Firing, Drawing 10520411, covering a detail of Drawing 10520406.

Characteristic	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Concentricity of large cavity with body	Gage
102	Concentricity of striker with body	Gage
103	Concentricity of cavity with body	Gage
104	Symmetry of pin relief holes with small cavity	
105	Symmetry of ball hole counterbore with ball hole	Gage
106	Diameter of striker	Gage
107	Diameter of body	Gage
108	Diamter of ball hole, max	Gage
109	Diameter of ball hole counterbore	Gage
110	Diameter of pin relief holes, min	Gage
111	Depth of ball counterbore, min	Gage
112	Length of striker, min	Gage
113	Location of ball hole counterbore	Gage
114	Metal defective	Visual
115	Protective finish inadequate or defective	Visual
116	Surface finish of body improper	Visual
117	Burr	Visual
118	Foreign matter present	Visual
Minor		
201	Diameter of large cavity	Gage
202	Diameter of small cavity	Gage
203	Diameter of ball hole, min	Gage
204	Diameter of pin relief holes, max	Gage
205	Depth of large cavity	Gage
206	Depth to bottom of small cavity	Gage
207	Depth of ball hole counterbore, max	Gage
208	Length of striker, max	Gage
209	Total length	Gage
210	Location of pin relief holes	Gage
211	Marking missing, incorrect, or illegible	Visual
212	Radius or chamfer missing or incorrect	Visual
213	Surface finish improper, except as otherwise classified	Visual

4.5.2.7 Spring, Initiator, Drawing 10520412, covering a detail of Drawing 10520406.

Critical

None defined

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Characteristic	Defect	Method of Inspection
Major		
101	Load at specified height	Gage
102	Closed length, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter, outside	Gage
202	Free length	Gage

4.5.2.8 Pin, Safety, Drawing 10522817, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Diameter of wire	Gage
102	Profile incorrect	Visual
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter present	Visual
Minor		
201	Diameter of head, outside	Measure
202	Length between high and low ends	Measure
203	Total length	Measure
204	Location of center loop	Measure
205	Radius of center loop incorrect	Measure

4.5.2.9 Filter, Drawing 10520414, covering a detail of Drawing 10520406.

Critical		
None defined		
Major		
101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of thread with center hole and outside diameter	Gage
104	Symmetry of holes with outside diameter	
105	Diameter of center hole	Gage
106	Diameter of holes	Gage
107	Depth of center hole	Gage
108	Depth of slot, max	Gage

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Characteristic	Defect	Method of Inspection
109	Location of holes	Gage
110	Angular location of holes	Gage
111	Nylon patch not properly located or fused	Manual/Visual
112	Number of holes incorrect	Visual
113	Metal defective	Visual
114	Protective finish, inadequate or defective	Visual
115	Burr	Visual
116	Foreign matter present	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of undercut	Gage
203	Diameter of body	Gage
204	Depth of slot, min	Gage
205	Total length	Gage
206	Length of body	Gage
207	Width of undercut, max	Gage
208	Width of slot	Gage
209	Location of slot	Gage
210	Marking missing, incorrect, or illegible	Visual
211	Chamfer missing or incorrect	Visual
212	Surface finish improper	Visual

4.5.2.10 Assembly, Drawing 10520406, less all drawings, except Drawings 10520410, 10520411, and 19060-3.

Critical		
1	Firing pin protrusion	Gage
2	Steel ball missing	Gage
Major		
101	Burr	Visual
102	Foreign matter present	Visual
Minor		
None defined		

4.5.2.11 Assembly, Drawing 10520406

Critical		
None defined		
Major		
101	Firing pin not cocked	Manual
102	Initiator pin not free to rotate	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Safety pin improperly inserted	Visual

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Characteristic	Defect	Method of Inspection
106	O-ring not lubricated	Visual
107	Marking missing, incorrect, or illegible	Visual
Minor		
201	Protective cap missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.12 Box, Fiberboard (Inner) Packing Ammunition, Drawing 8830984

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Box damaged	Visual
203	Sealing strip missing, improperly applied	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Cut, damaged, or improperly sealed bag	Visual

4.5.2.13 Box, Packing Ammunition, Drawing 8830985

Critical

None defined

Major

None defined

Minor		
201	Contents loose	Manual
202	Strapping missing, loose, broken, or improperly assembled	Manual/Visual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.14 Box, Fiberboard (Outer) Alternative, Packing, Ammunition, Drawing 9241155

Critical

None defined

Major

None defined

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Characteristic	Defect	Method of Inspection
Minor		
201	Contents loose	Manual
202	Box broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each chamber and cap subassembly shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with the requirement of 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Firing pin release. The initiator pin of each partial assembly consisting of all parts, except the safety pin and chamber, shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirement of 3.5 shall be cause for rejection of the subassembly.

4.5.3.4 Initiator pin security. The initiator pin of each subassembly consisting of all parts, except the chamber, shall be subjected to the initiator pin security test. Failure of any partial assembly to comply with the requirement of 3.6 shall be cause for rejection of the subassembly.

4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any sample in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit in the inspection sample to comply with the requirements of 3.9 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to the application of protective finish, the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with the requirement of 3.4.

4.6.3 Firing pin release. The initiator pin of each partial assembly consisting of all parts except the safety pin and chamber shall be subjected to the firing pin release test. Failure of any partial assembly to comply with the requirements of 3.5 shall be cause for rejection of the subassembly. To achieve the pull force requirements of 18-25 pounds specified in paragraph 3.5, selective reassembly may be necessary (i.e., changing initiator pins) for those assemblies which are initially out of range.

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4.6.4. Initiator pin security. The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with the fixture which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for 4 full passes.

4.7 Inspection of packaging. The sampling and inspection of the packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING.

5.1 Packing and marking. Packing and marking shall be levels A or B as specified by the contracting activity:

Level A	In accordance with Drawings 8830984 and 8830985
Level B	In accordance with Drawings 8830984 and 9241155

5.2 Air Force packaging. For Air Force packaging use TPO 00-845-1059.

6. Notes.

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

c. If first article inspection is required and Government inspection facility (see 3.1, 4.4, and 6.3)

d. The responsible inspection activity (see 4.1)

e. Level of marking and packing (see 5.1).

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, and 1 set of unassembled parts without plating or final protective finish. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested

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by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definition.

6.4.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy-OS
Air Force-99

Preparing activity:
Navy-OS
(Project 1377-N949)

Review Activity:
Air Force-70

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